Pavan Dev Yarakala

co24btech11025@iith.ac.in | pavandevyarakala@gmail.com | +91 9989031979

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	5.58
Narayana	Class XII (Telangana state board of intermediate education)	2024	91.20%
Zee high school	Class X (CBSE)	2022	89.00%

SKILLS

Programming Languages: Python, C, C++

Development : Git , GitHub , Firebase ,Android Studio , VS Code , Google Cloud Platform (Fundamentals), AI-Assisted Software Development, UI/UX Development , Debugging & Testing

Libraries & Tools: NumPy, Pandas, Matplotlib

Productivity & Collaboration: Microsoft PowerPoint, Microsoft Word, Google Docs, Google Sheets

RELEVANT COURSES

Core Courses: Introduction to Programming , Data Structures and their Applications, Operating Systems , Fundamentals of Scientific Computing , Computer-oriented Numerical Methods

Math Courses: Calculus I, Calculus II, Probability , Intro to Statistics, Elementary Linear Algebra, Differential Equations, Complex Variables

Business & Communication - Introduction to Entrepreneurship , Communication Skills

PROJECTS

Re:Focus - Android Digital Wellbeing & Short-Form Content Tracking Application

Tech Stack: Kotlin, Jetpack Compose, Android AccessibilityService, Coroutines/Flow, Room, Firebase Authentication, Firebase Firestore, Dagger-Hilt, MVVM, HTML5, CSS3, JavaScript

- **Developed the project using an AI-assisted engineering workflow**, leveraging AI for code generation, debugging, architecture iteration, and UI development while implementing and validating the underlying Android systems.
- **Engineered** a real-time Instagram Reels and YouTube Shorts detection engine using Android AccessibilityService, BFS-based accessibility-tree traversal, semantic fingerprinting, spatial filtering, and directional scroll detection to prevent duplicate and false-positive counts.
- **Designed** a high-performance event-processing pipeline using Kotlin Coroutines, buffered Channels, 50ms event debouncing, and Mutex-based synchronization to handle rapid accessibility events while preventing race conditions.
- **Built** a reactive Jetpack Compose UI using MVVM and Dagger-Hilt, including a system-level WindowManager floating counter and real-time daily usage controls with configurable limits and Strict Mode enforcement.
- **Implemented** an offline-first data architecture using Room/SQLite and DataStore, with Firebase Authentication and Firestore synchronization featuring custom conflict resolution for historical and current-day usage data.
- **Built and deployed** the official responsive product website using HTML5, CSS3, and JavaScript ; deployed via Firebase Hosting and distributed versioned Android APK releases through GitHub Releases.

GitHub: https://github.com/pavandevyarakala-create/ReFocus | **Live Website:** https://refocus-504119.web.app/

Inventory Management System

Built a responsive inventory management platform featuring product and category management, stock-in/stock-out

tracking, searchable inventory, and real-time dashboard analytics. Designed to streamline warehouse operations and improve inventory accuracy through an intuitive user experience.

Github: https://github.com/pavandevyarakala-create/Inventory

POSITIONS OF RESPONSIBILITY

Head of Infrastructure and Security Domains of Elan and nVision 2026-2027

Infrastructure: Leading infrastructure planning, logistics, venue readiness, and cross-functional coordination to support the execution of one of IIT Hyderabad's largest annual festivals.

Security: Coordinating with institute security, administration, and organizing teams to plan access management, safety protocols, and on-ground security operations for a large-scale, multi-day campus event.

Coordinator, Security Domain | Elan & nVision 2025-26

Coordinated with institute security, organizing teams, and volunteers to support secure operations and smooth execution of one of IIT Hyderabad's largest student-run festivals.

Coordinator, Events Domain | Milan 2025-26

Coordinated logistics and on-ground execution across cultural, technical, and sports events, collaborating with multiple teams to ensure seamless operations throughout IIT Hyderabad's inter-hostel competitions.

Coordinator, Finance Domain | Ek Bharat Shreshtha Bharat Club (2025-26)

Coordinated budget planning, expense management, and financial documentation while collaborating with organizing teams to support successful club events.

Coordinator, Logistics Domain | Sunshine Management

Managed event logistics, vendor coordination, inventory movement, and operational support to enable seamless execution of multiple events.

Volunteer | Office of Career Services (OCS)

Supported campus recruitment activities by assisting with candidate coordination, event logistics, and seamless execution of placement drives.